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Title: The Birth of Machine Mind: A Child-Inspired Neural Network

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ABSTRACT

This paper introduces a novel AI model inspired by the cognition of a newborn, aiming for self-learning without preloaded datasets. The system uses a 6-billion-neuron structure with customized activation functions, mimicking the adaptability of the human brain. Unlike traditional models, this approach processes real-time sensory data, enabling emergent cognition. We analyse its performance through mathematical formulations and experimental comparisons, highlighting its advantages over existing AI systems. The results suggest that independent AI thinking is feasible, with promising applications in autonomous decision-making and human-AI collaboration.

Keywords— Brain, AI, Self-Think, Neural Network, Artificial brain

I.INTRODUCTION

The modern world has developed massive AI models that rely on extensive datasets collected globally to generate unique and reliable insights. However, these models function purely on pre-existing data and predefined algorithmic rules. If asked something outside their dataset, they either fail to respond accurately or generate answers that are not truly their own.

This does not mean that AI models are useless; rather, they do not fully embody the meaning of Artificial Intelligence as the term suggests—machines exhibiting intelligence through artificial means. Instead of truly "thinking," current AI models apply statistical and algorithmic operations on pre-stored data to generate results that appear unique and realistic. However, this approach remains data-driven rather than naturally intelligent.

To illustrate my concept, consider a newborn baby. At birth, the baby does not have pre-existing thoughts or knowledge. Yet, as it experiences the world—seeing its mother for the first time, hearing sounds, and touching objects—its brain begins to form associations. Suppose a baby sees an apple for the first time. Without any prior data, the baby experiments—touching, tasting, and observing. Over time, the brain categorizes the apple as something edible. Now, if the baby encounters a red ball, its brain, based on shape similarity, might initially attempt to treat it as an apple. However, upon trying to bite it, the baby realizes it is not edible. This trial-and-error learning process is how human intelligence develops—by interacting with the environment, making mistakes, refining knowledge, and adapting behaviours.

Inspired by this natural cognitive development, I propose a brain-like AI system that does not rely on preloaded datasets but rather learns dynamically in real-time. The system is designed to replicate the lobe-based architecture of the human brain, where different lobes perform specific tasks but work together to process experiences, test scenarios, and refine knowledge.

Unlike conventional AI, which depends on existing datasets, my model focuses on visual and auditory inputs to establish relational understanding and generate multiple scenario-based outputs. The system continuously refines itself by testing, validating, and differentiating between correct, incorrect, and expected outcomes, much like a human brain. While the initial implementation may have unstructured, unpredictable, or unreadable outputs, the approach follows the true essence of Artificial Intelligence—an AI system that learns, evolves, and adapts without relying on predefined datasets.

In this paper, I present a detailed theoretical framework along with practical implementations to demonstrate the feasibility and usability of this AI brain model, paving the way for real-time intelligence beyond conventional dataset-driven approaches.

II.METHODOLOGY

a). Human Child Brain Structure to AI Mimic:

The human brain is composed of multiple lobes that perform specialized functions. To mimic this biological intelligence, the artificial brain is designed with neurons that form a structured, interconnected system. The entire system follows a logical communication model, where data flows between different brain regions, just like in a real human brain.

In biological brains, the frontal, temporal, parietal, occipital, and cerebellum lobes handle different tasks. Similarly, in the artificial brain:

✓ Each lobe consists of 8 functional modules, each performing a unique role (mention in Fig no-01).

✓ Each lobe is connected through three communication lines (mention in Fig no-01):

i. A main data line that transmits information across all lobes (mention in Fig no-02).

ii. Two internal interconnection lines that allow communication within the lobe itself.

✓ Each lobe has a dedicated processor that manages internal operations and ensures smooth data flow (mention in Fig no-01).

✓ A central processor oversees all lobes, acting as the main brain processor, handling input-output communication and decision-making (mention in Fig no-02).

This structured brain-inspired architecture enables the system to function dynamically, processing real-time data and adapting like a human brain.

i). Lob Design:

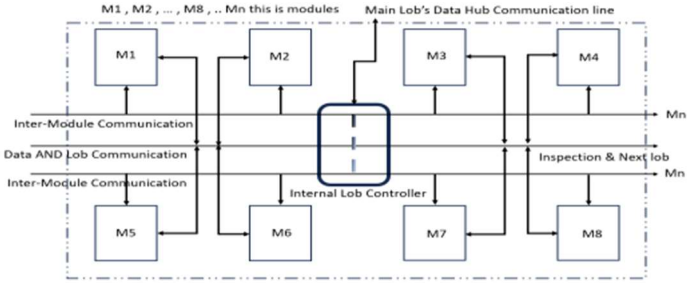


Fig no – 01

Assume this any lob their in have 8 module called as M1, M2, M3,,…Mn and Data line and Internal communication line with have internal lob controller and their external connected to Central Main Hub.

ii). Entire Brain Design:

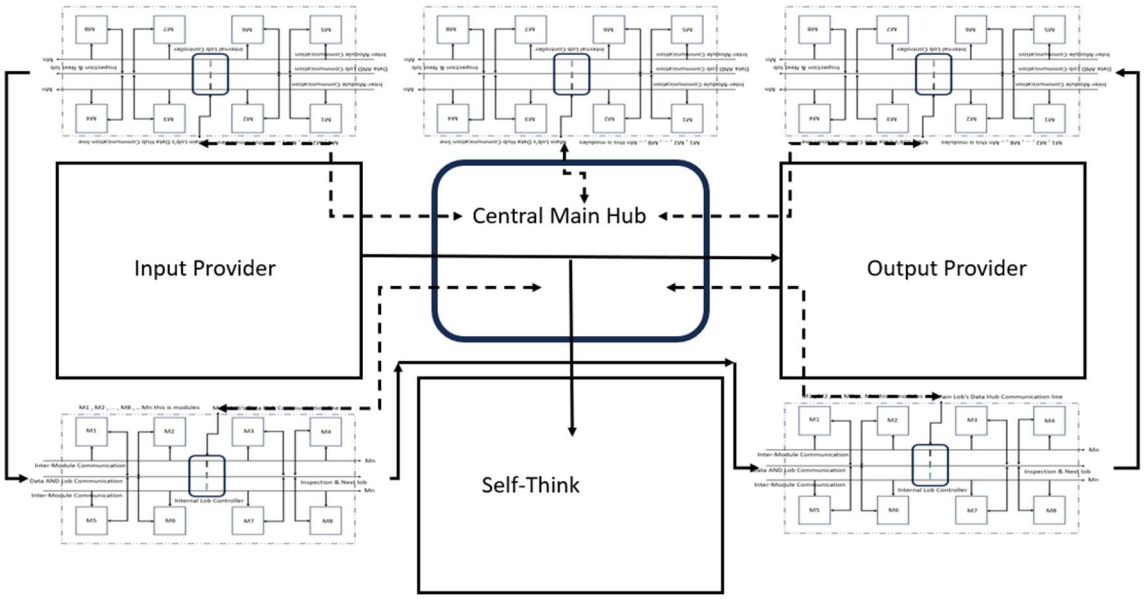


Fig no – 02

Above diagram contain 5 lob design, input and output provider and Central Main Hub to handle everything of entire brain. And self-think this is main part to be main motive of this paper.

b). Mathematical Concept for Neural Activation, System Main Element and Self-think:

Mathematical Perspective of artificial brain based on various section have own formulas to be easily identifies the utilization of system to using simple but formulas as well as neuron working on the custom activation function working flow this are the base of Artificial brain.

Basically, Mathematical it's represented the brain entire and internal numerical computation understand and helping to determine the influence factor and update.

i). Neuron activation custom function formula:

$$\sqrt{\left[\frac{(a^2+c^2)}{2}\right]-\left[\frac{(c-a)}{2}\right]}$$

This is LNR (Linear Number Relation) formula to have potential to make linear and without linear output production as per stages of hidden layer and output layer with have more mathematical logic to make this stable and useable but this not easy to cover in this paper.

ii). System Main Element Formula:

$$\frac{\sum_{i=no\ of\ lob's\ utlization}(L_{\ u})_{\ i}}{\mathrm{DL}_{\ u}}$$

This is Entire brain handling data , lob and Input , Output controlling and handling critical element in the brain have own various approaches to design and make it more efficient as per above formulas just introduce about critical element utilization of influence factor how to find out this is not means this formula are have potential to cover this critical element and this not possible in this paper.

iii). Self-think:

$$\frac{Multiple(Entity*Influence)}{Single(Entity*Influence)}$$

This if Self-Think formula but this is just critical approach in single step to brain can act like self-think in details formulas are more critical and under progress to not have stabilize forms and verified with proof.

This are the critical formula to showing the brain processing and actual operation but this are not have potential to defined entire brain structure because entire brain have various part each parts and their own process and each perspective have own formula and theory and this not possible to discuss to here without any proof.

c). Development:

The transformation of a newborn baby's brain into an artificial brain is achieved using neurons and various AI algorithms. The system processes real-time data to recognize key logical elements, simulating the way human cognition develops. Each element is implemented through neurons, systematically coded for execution, ensuring the model behaves similarly to a child's learning process.

This section does not provide extensive details, as the full research, implementation, and optimization will be presented in future papers focusing on the practical aspects of this system.

d). Testing and Validation (Future Scope):

The proposed artificial brain model is still under development and testing will play a crucial role in improving its accuracy and efficiency. The initial validation approach will focus on evaluating the stability of the activation task, neuron decision-making, and inter-lobe communication efficiency. In addition, self-thinking evaluation testing will be conducted based on different real-time inputs.

Future experiments will compare the decision-making process of the model with that of a traditional AI model to demonstrate its ability to operate without a preloaded dataset. The response time, accuracy, and adaptability of the system will be key performance indicators. Further refinements will be made based on empirical observations and real-time processing results.

III. RESULT AND DISCUSSION

a). Result:

The proposed artificial brain model is designed to work by making real-time observations and dynamically processing data, rather than relying on preloaded datasets. The expected outcome of this research is to successfully mimic the cognitive learning process of a child's brain, where the system observes, processes, and adapts to its environment over time.

Unlike traditional AI, which relies on large datasets, this system aims to develop self-learning behavior using neuron-based activation functions. Initial theoretical analysis suggests that the model should be capable of:

i). Real-time processing without pre-nourished knowledge
ii). Self-generated learning through observed interactions
iii). Scalable neural network adaptation based on activation responses.

Discussion:

This approach presents a radical shift in AI design, prioritizing human-like cognitive functions over traditional data-driven learning. While existing AI models rely on training data, this research introduces a self-thinking, adaptive system that continuously refines its responses without a preloaded dataset.

IV.CONCLUSION AND FUTURE WORK

Conclusion:

This research presents an innovative approach to artificial intelligence, which aims to develop an AI system that thinks for itself, inspired by the cognitive structure of the human child brain. Unlike traditional AI models that rely on preloaded datasets, this system processes real-time data through custom neuron activation functions so that it can dynamically learn, adapt, and produce output.

The proposed framework represents a breakthrough in AI development by enabling autonomous learning, logical decision-making, and interaction without the need for prior training data. The mathematical modeling and neural communication structures described in this study lay the foundation for further refinement and practical implementation.

However, this research is in its early conceptual stage and many challenges remain, in particular stabilizing neuron-based activation, optimizing response time, and ensuring a controlled learning process.

Future Work:

Future iterations of this model will focus on **enhancing the efficiency and accuracy of real-time learning**. Key areas of improvement include:

- **Experimental Testing & Implementation** – Developing an initial prototype and conducting test cases to validate the theoretical framework.
- **Optimization of Neural Activation Functions** – Refining the mathematical model to ensure stable and meaningful AI responses.
- **Benchmarking Against Existing AI Models** – Comparing performance with traditional machine learning models to evaluate improvements.
- **Hardware Integration** – Exploring compatibility with neuromorphic chips and advanced computational architectures for better real-time processing.
- **Ethical & Safety Considerations** – Ensuring AI decision-making remains controlled, unbiased, and aligned with human expectations.

This research lays the **groundwork for the next generation of AI**—an AI that does not merely mimic intelligence but **thinks and learns like a human child**, marking a significant step toward **true artificial cognition**.

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